Where Inventions Came From

A map of invention from the first stone tools to the present,
from Wikipedia and Wikidata

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Abstract

Where did inventions come from, and how did the centre of invention move? I combine three open sources into one map. A hand-checked table, built from the text of nine English Wikipedia pages, places and dates 115 early innovations at 155 places, keeping date ranges and several places wherever the pages give them. The 314 entries since 1500 in Wikipedia's Timeline of historic inventions extend this curated list to the present. And 1,443 inventions from Wikidata, filtered from 24,186 items with an inventor or a date of invention, add a much larger but much less even layer, placed by where they were made, the inventor's birthplace, or the country of origin. Six innovations arose independently in several places on the pages' own account, among them farming (11 centres), writing (4) and zero (3). Averaging each period's locations, the centre of invention sits in Africa in the deep past, in West Asia from 10,000 BC to AD 1 (where 51% of the landmarks of 5000 to 2000 BC came from), in East Asia from AD 1 to 1500 (50%), in Europe from 1500 (all 16 landmarks of 1500 to 1700), and in North America from 1950 to 2000 (55%). The open record is lopsided: 91% of Wikidata's mapped inventions are in Europe or North America, regions with 14% of today's population, and 83% are credited to named individuals. The map describes the written record of invention, not invention itself.

Keywords: history of technology; invention; independent invention; domestication; Wikipedia; Wikidata; geographic bias; mapping

1 Introduction

Inventions are usually told as stories of one place and one person: Gutenberg in Mainz, Bell in Boston, Edison in New Jersey. Historians of technology have long resisted that framing. Many of the most important innovations, from farming to writing, were made independently more than once (Diamond, 1997; Larson et al., 2014); for most of recorded history the most inventive societies were in West Asia and China (Needham, 1954; Mokyr, 1990); and much of what matters is the use and spread of old things rather than the moment of first invention (Edgerton, 2006).

This paper asks a simple descriptive question with open data: if every invention is put back where it came from, what does the map look like, and how did its centre move over 3.3 million years? It pairs the analysis with an interactive atlas (https://maps.tristanhotham.com/inventions-map/) with a time slider, filters for eleven kinds of invention and a note on how each point was placed.

The answer depends heavily on the source. English Wikipedia and Wikidata (Vrandečić and Krötzsch, 2014) are written mostly by people in Europe and North America, about things they know, and their geography is famously uneven (Graham et al., 2014). They also frame invention as the act of a named individual, which suits patents and fails for the domestication of wheat. So the paper treats the sources' biases as a result in their own right.

Table 1: From query to map: the Wikidata layer.

Step	Items
Rows returned by the query	36,849
Items (inventor or discoverer, or a date of invention or discovery)	24,186
dropped: class is not a technology (place, person, artwork, find, idea...)	13,616
dropped: no English label	2,619
dropped: no technology class or word, or dropped by hand	5,524
Kept as inventions	2,334
with 5 or more sitelinks (the Wikidata layer)	1,443
placed at a location of invention or discovery	52
placed at the inventor's birthplace	949
placed at the country of origin	21
not placed	421
with a date	578

2 Data

Early innovations. Nine English Wikipedia pages (Wikipedia contributors, 2026), saved in 2026: Timeline of historic inventions, Timeline of prehistory, Neolithic Revolution, Domestication, History of writing, Wheel, Bronze Age, List of Indian inventions and discoveries, and Islamic Golden Age. From their text I wrote down 155 rows for 115 innovations from the first stone tools to Gutenberg's press (Table 5).

The timeline since 1500. All 314 dated entries from 1500 onwards in the Timeline of historic inventions, from rifling in 1540 to large language models.

Wikidata. A query through QLever (Bast and Buchhold, 2017) returned 24,186 items with an inventor or discoverer (P61) or a time of invention or discovery (P575), excluding taxa, astronomical objects and chemical compounds: their English labels, number of sitelinks, dates, classes (P31), inventors with birthplaces and citizenship, locations of discovery (P189) and creation (P1071), and countries of origin (P495).

Geography. Natural Earth (Natural Earth, 2026) countries, populated places and population estimates. Wikidata is CC0, Natural Earth public domain; Wikipedia text is CC BY-SA 4.0, so the two tables built from it, which quote short passages, are shared under that licence.

3 Methods

3.1 The hand table of early innovations

Each row gives an innovation, a place as the page names it, a date or date range as the page gives it, a category, the page, and the passage the row rests on. A script checks every passage against the saved page text. Dates in thousands or millions of years ago are converted with 1950 as the present. Where a page names several places, each becomes a row of one of three kinds: *independent* origins, where the page says the innovation arose separately (farming, livestock, bronze, writing, zero and the suspension bridge); *candidates*, where it gives alternatives (glass in "either Mesopotamia or Ancient Egypt"); and *finds*, several early finds without a verdict (the wheel). Coordinates are mine and rough, at one of four levels: a site (43 rows), the middle of a region (78), of a country (27) or of a continent or wider (7). The page gives no place for the control of fire; it is shown in

Africa, where the same page places the period's other tools, with the widest uncertainty.

3.2 The timeline since 1500

Each entry is placed by a place it names (a city, laboratory or company: 70 entries), else by the birthplace of a person it names who is an inventor in the Wikidata extract (112), else by a country or nationality it names (3). The remaining 120 (38%) name only a person or firm not in the extract; I placed these by hand at the centre of the country where that person or firm did the work, from general knowledge rather than the page, and flag them. 9 entries (rifling, the flintlock, the pocket calculator and others) could not be placed.

3.3 Filtering Wikidata

Most of the items are not inventions. I dropped items whose classes mark natural objects, places, people, artworks, archaeological finds, diseases, theories, games and sports, dishes and brands (13,616 items), and items with no English label (2,619). I kept items with a class that is a kind of technology (weapon, vehicle, machine, instrument, medicine, software, writing system, material and so on) and, because Wikidata often describes general kinds of thing such as "refrigerator" with no instance-of class at all, unclassed items whose name contains a technology word. I then read every kept item with ten or more sitelinks and every dropped item with twenty or more, and overrode about three hundred decisions by hand, dropping scientific discoveries (electricity, the electron), effects, statistical methods and foods, and rescuing items such as photography, nylon and the QR code. This leaves 2,334 items, of which 1,443 have 5 or more sitelinks, the number of Wikipedia and other Wikimedia pages about them, used throughout as a measure of fame. Categories come from the name, else from the class.

3.4 Placing Wikidata inventions

Each invention is placed at its location of discovery or creation if that has coordinates (52); else at the birthplace of its first-listed inventor, ordered by Wikidata identifier (949); else at the Wikidata coordinate of its country of origin (21). 421 have none of these. The inventor rule maps where inventors were born, which is often not where they worked: the method is recorded for every point and shown on the map.

3.5 Countries, regions, cities and the centre of invention

Points are given the present-day country they fall in and one of nine world regions; Siberia and the Eurasian steppe in the hand table are counted with Central Asia. City areas group Wikidata points (of the first two kinds) within 25 km of a Natural Earth place of 150,000 people or more. The centre of invention for a period is the weighted mean latitude and mean longitude of its points, each counted once and candidate places sharing one count. A mean on the globe would be pulled towards the pole when Europe, America and Japan are averaged, so the flat mean is easier to read; either way a centre can fall in the sea, and is a balance point rather than a place. The main series, the *landmark list*, uses the hand table before 1500 and the timeline from 1500; Wikidata is a second series from 1700, where it has at least fifteen dated, placed inventions per period.

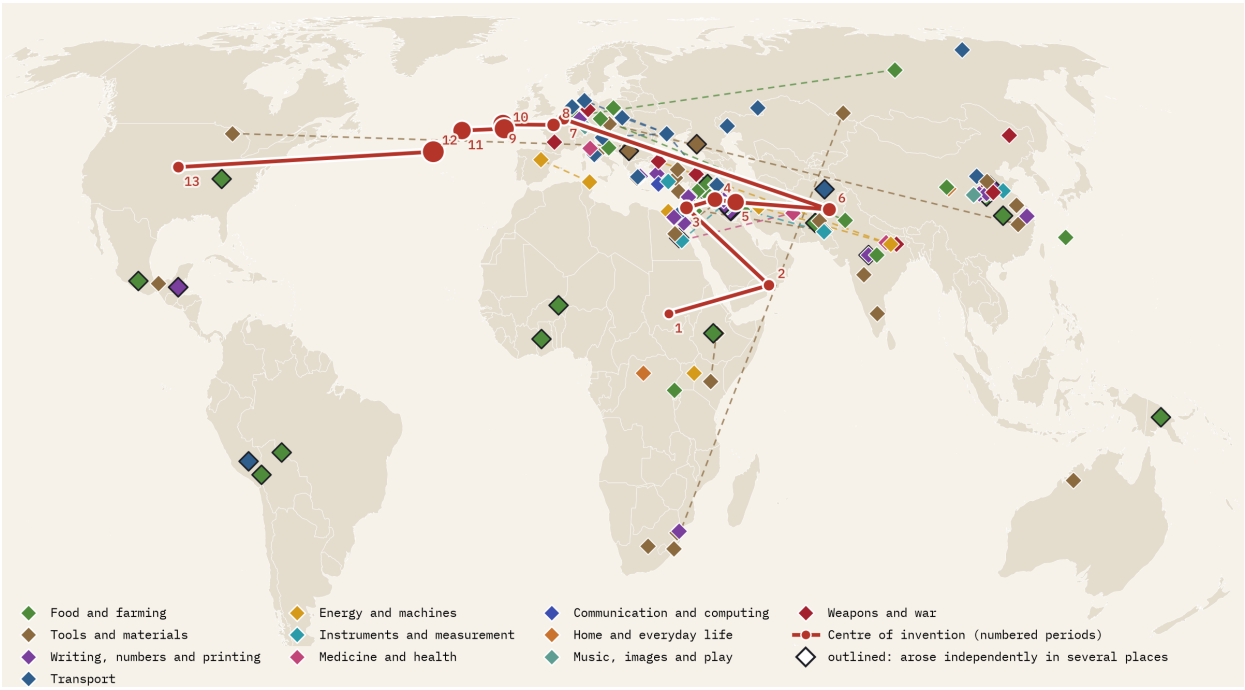


Figure 1: The 155 places of 115 early innovations, coloured by kind; outlined diamonds arose independently in several places, dotted lines join early finds or candidate places. Red: the centre of invention of the landmark list, numbered by period as in Table 3. Equal Earth projection.

4 Results

4.1 The early map and independent origins

Figure 1 maps the hand table. The oldest rows are East African stone tools (3.3 million years ago). Six innovations arose independently in several places according to the pages (Table 2). Farming is the clearest case: eleven centres from the Fertile Crescent (10,000 to 8000 BC) and the Yangtze basin to the New Guinea Highlands, southern Mexico, the Amazon basin, the Andes, three African centres and eastern North America. Writing arose four times, in Mesopotamia, Egypt, China and Mesoamerica; zero three times, in Babylon, among the Maya and in India. 7 innovations have candidate places and 9 several early finds.

4.2 The moving centre of invention

Table 3 and Figure 2 show where each period's landmarks came from. In the two oldest periods the largest share is sub-Saharan Africa. From 10,000 BC to AD 1, West Asia and North Africa lead: 44% of the landmarks of 10,000 to 5000 BC, 51% of those of 5000 to 2000 BC. From AD 1 to 1500, East Asia, in practice China, has 50% of 26 landmarks: paper, woodblock printing, porcelain, the banknote, gunpowder, movable type. The page lists only a handful of medieval European inventions before Gutenberg, which is itself a choice of the page.

From 1500 to 1700 every landmark on the list is European, and Europe holds 79% from 1800 to 1850 and 77% from 1850 to 1900, with North America at 21% in the latter. North America first leads from 1950 to 2000 (55%), and all 16 entries since 2000 are American. East Asia's return is modest on this list: 12% from 1950 to 2000. The Wikidata series agrees: 96% European from 1800 to 1850, and North America ahead from 1950 to 2000 (45% against Europe's 40%). The centre therefore travels from East Africa through the Levant, Mesopotamia and Iran towards South Asia,

Table 2: Innovations that arose independently in several places, according to the pages.

Innovation	Places	Where
Farming	11	Fertile Crescent (wheat, barley, lentils); Yangtze basin, China (rice); Yellow River basin, China (millet); Kuk Swamp, New Guinea Highlands (taro, bananas); Southern Mexico (maize, squash, beans); Southwestern Amazon basin (cassava); Lake Titicaca, Andes (potatoes); Ethiopian highlands (coffee, teff, ensete); Sahel (sorghum, pearl millet); West Africa (African rice, yams, oil palm); Eastern North America (sunflower, sumpweed, goosefoot)
Livestock	3	Fertile Crescent (goats, sheep, pigs, taurine cattle); Baluchistan, Pakistan (humped zebu cattle); East Asia (pigs)
Bronze	3	Pločnik, Serbia (tin bronze foil, dating disputed); Maykop culture, North Caucasus (arsenical bronze); Sumer, Mesopotamia
Writing	4	Uruk, Mesopotamia (cuneiform); Abydos, Egypt (hieroglyphs); Late Shang China (oracle bone script); Maya region, Mesoamerica (Maya script)
Zero	3	Babylon; Maya region, Mesoamerica; India (Brahmagupta)
Suspension bridge	2	Pre-Columbian South America; Hindu Kush (Afghanistan and Pakistan)

Table 3: The centre of invention: mean latitude and longitude of each period's landmark inventions, and the region with the largest share.

	Period	Items	Lat, lon	Centre falls	Largest share
1	3.3 million to 100,000 years ago	9	12.4, 28.4	Sudan	Sub-Saharan Africa (67%)
2	100,000 to 12,000 years ago	15	17.5, 52.8	Oman	Sub-Saharan Africa (33%)
3	10,000 to 5000 BC	25	31.8, 34.7	Israel	West Asia and North Africa (44%)
4	5000 to 2000 BC	35	33.4, 42.2	Iraq	West Asia and North Africa (51%)
5	2000 BC to AD 1	45	32.9, 47.4	Iran	West Asia and North Africa (41%)
6	AD 1 to 1500	26	31.5, 70.9	Pakistan	East Asia (50%)
7	1500 to 1700	16	50.0, 4.0	France	Europe (100%)
8	1700s	24	48.7, 1.0	France	Europe (92%)
9	1800 to 1850	62	48.8, -13.3	open sea, nearest land Ireland	Europe (79%)
10	1850 to 1900	62	47.8, -12.8	open sea, nearest land Ireland	Europe (77%)
11	1900 to 1950	51	47.4, -24.6	open sea, nearest land Portugal	Europe (63%)
12	1950 to 2000	74	43.0, -31.6	open sea, nearest land Portugal	North America (55%)
13	2000 to 2026	16	39.8, -98.6	United States	North America (100%)

jumps to northern France, and drifts west across the Atlantic to the United States.

4.3 Inventions per century

Both lists accelerate sharply (Figure 3). Wikidata has 177 dated inventions from the nineteenth century and 282 from the twentieth, but 34 from before 1500, and 84% of its dated inventions are from 1800 or later. Part of this is the real acceleration of invention; part is that recent inventions

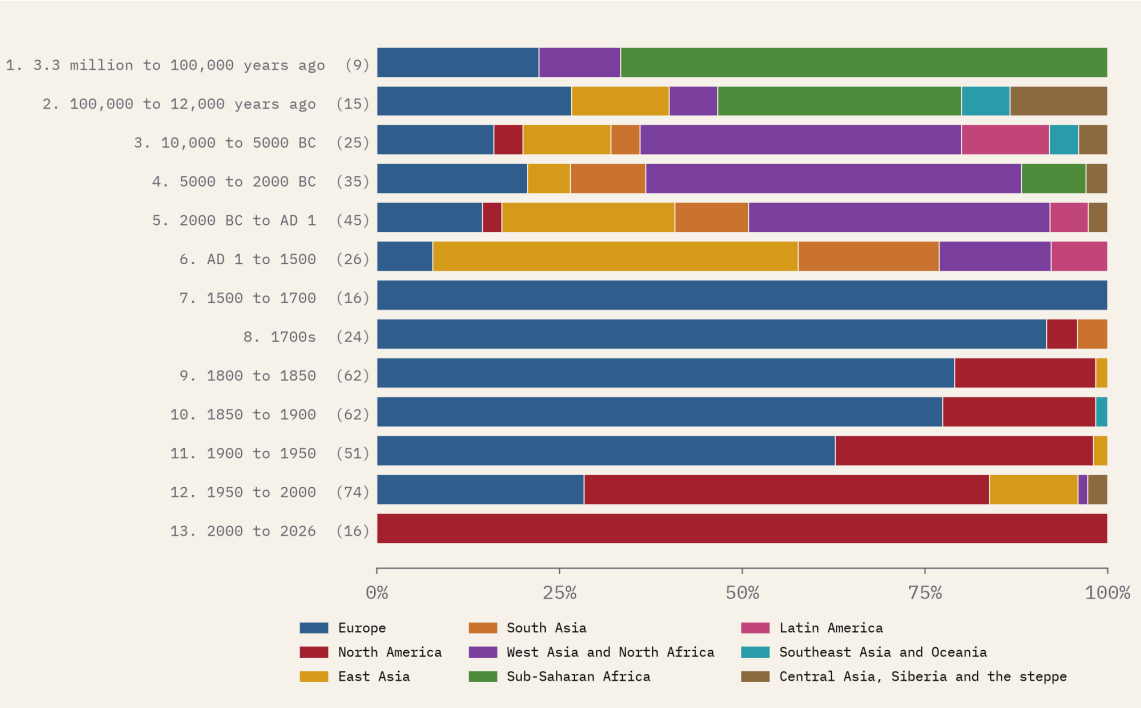


Figure 2: Share of each period's landmark inventions by world region (hand table before 1500, Wikipedia timeline from 1500; number of items in brackets).

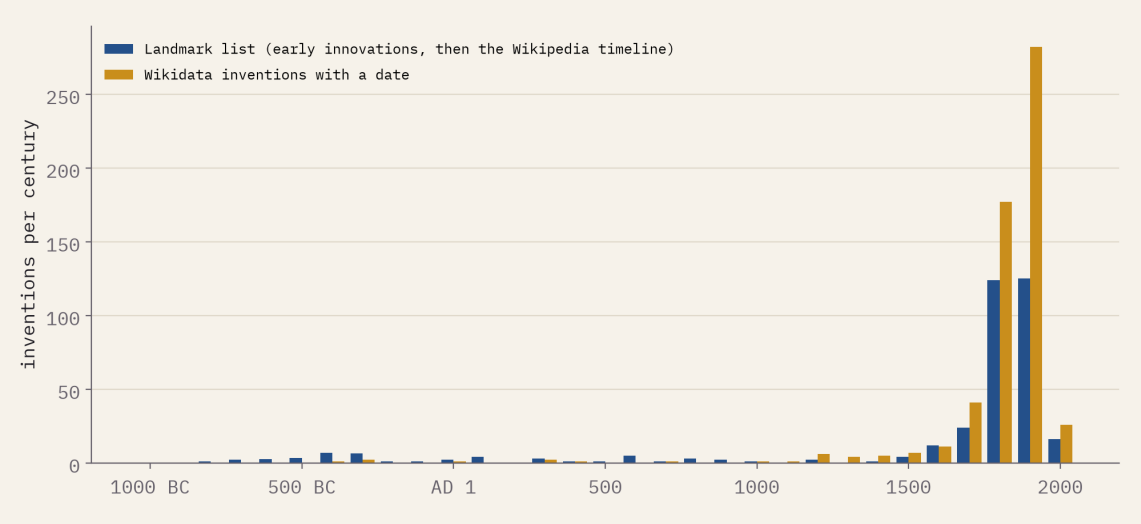


Figure 3: Inventions per century, 1000 BC to the present: landmark list and dated Wikidata inventions. The last bar is 2000 to 2026.

are documented, patented and attached to named people.

4.4 The Wikidata layer and the most inventive cities

Of the 1,443 Wikidata inventions, 1,021 (71%) can be placed and 578 (40%) have a date (Figure 4). By country the United States leads (260), then the United Kingdom (156), France (122) and Germany (120). By city area London leads with 47, then New York (37), Paris (27) and Vienna (16) (Table 4); most are inventors' birthplaces.

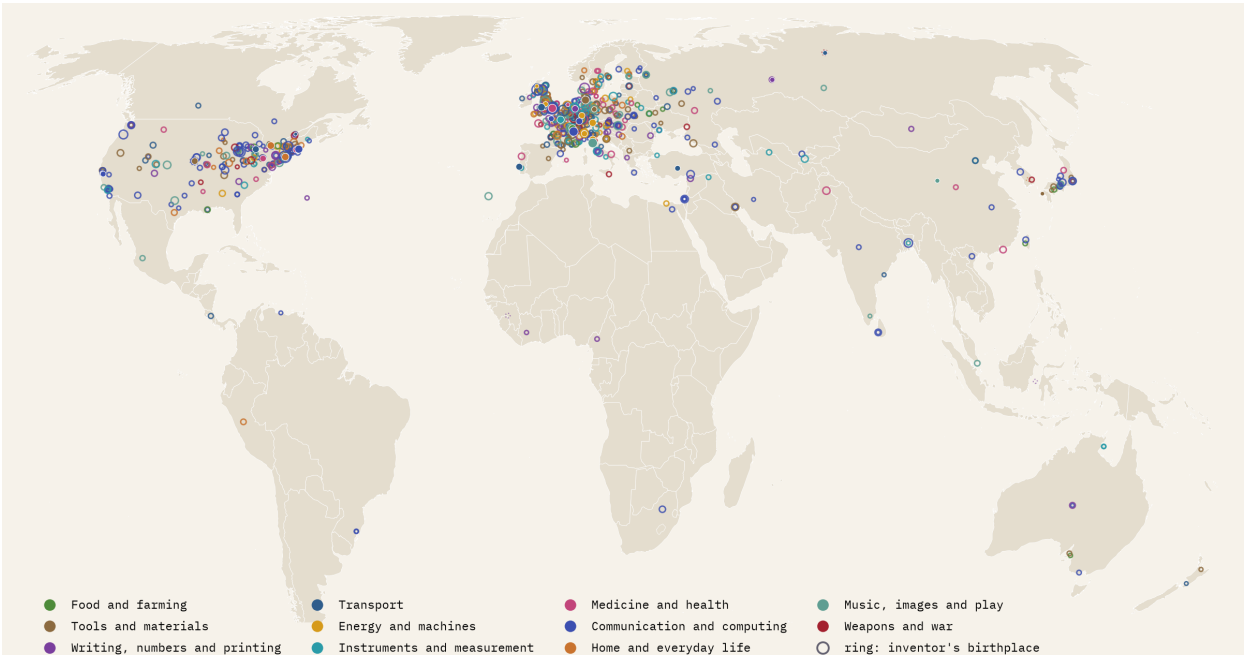


Figure 4: Wikidata inventions with 5 or more sitelinks that can be placed (1,021), sized by sitelinks and coloured by kind; filled where Wikidata gives a place of invention, rings where the point is the inventor's birthplace.

Table 4: City areas with the most Wikidata inventions: all, inventor born there, made or discovered there, and the best known.

City	Country	All	Born	Made	Best known
London	United Kingdom	47	39	8	computer, HTML, web browser
New York	United States	37	32	5	Internet, escalator, Ethernet
Paris	France	27	23	4	animation, internal combustion engine, sonar
Vienna	Austria	16	16	0	triode, Glock, Glock 17
Chicago	United States	12	12	0	mobile phone, Hammond organ, WD-40
Boston	United States	10	9	1	Morse code, lightning rod, bolometer
Edinburgh	United Kingdom	9	9	0	telephone, cloud chamber, Rankine cycle
Tokyo	Japan	8	8	0	QR code, karaoke, marker pen
Munich	Germany	8	7	1	diesel engine, variometer, Volksempfänger
Philadelphia	United States	8	8	0	Internet Message Access Protocol, iron lung, LM317
The Hague	Netherlands	7	7	0	capacitor, compact disc, Bluetooth
Luton	United Kingdom	7	7	0	Linear B, Bessemer process, octant
Berlin	Germany	7	7	0	thermite, breadth-first search, Feistel cipher
St. Petersburg	Russia	7	7	0	stop-motion, theremin, drogue parachute
San Francisco	United States	7	7	0	Network Time Protocol, jackhammer, hash tree

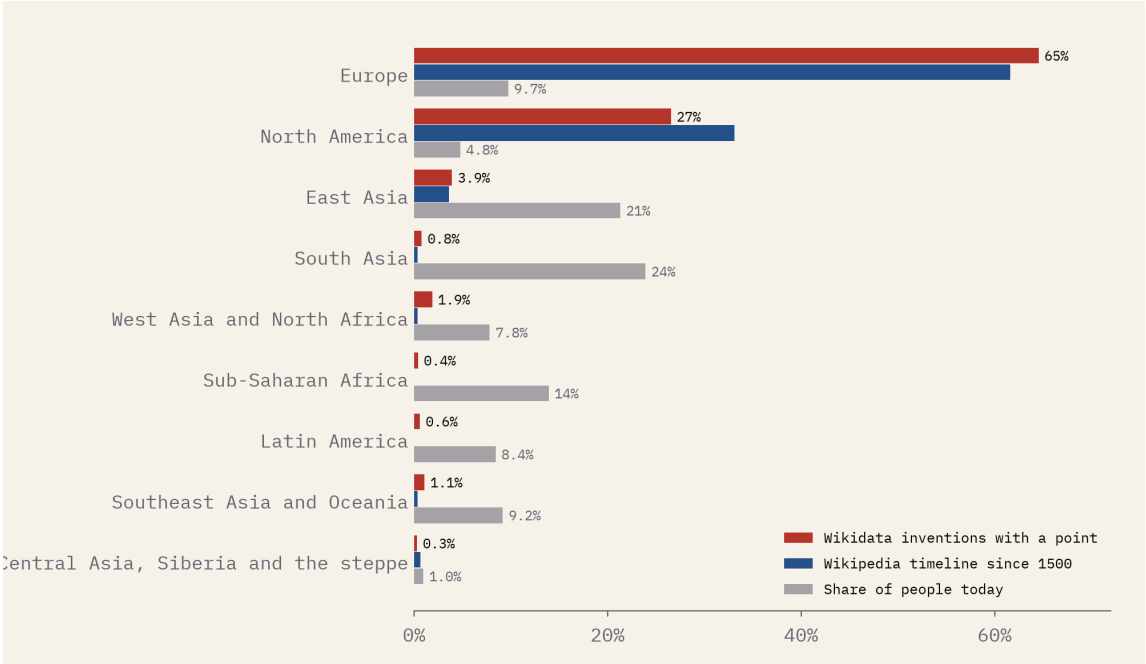


Figure 5: Share of mapped inventions (Wikidata; Wikipedia timeline since 1500) and of present-day population, by region.

4.5 Who gets the credit

Figure 5 sets each region's share of mapped inventions against its share of today's people. Europe and North America hold 91% of the Wikidata inventions and 95% of the timeline since 1500, but 14% of the population; South Asia has 0.8% of Wikidata's inventions and 24% of the people, sub-Saharan Africa 0.4% and 14%. 83% of the Wikidata inventions have a named inventor; the hand table, by contrast, names almost no one. Individual credit throws up oddities: Wikidata's only inventor of the car is Ferdinand Verbiest, who built a small steam vehicle in China, and it credits gunpowder to Berthold Schwarz, a legendary German monk, while the Wikipedia timeline dates it to ninth-century China.

5 Discussion

Two stories sit in the same map. One is the familiar history of technology: an early world of many independent beginnings, a long period in which West Asia and then China led, and a modern period in which Europe, then North America, took over. The landmark list, curated and short as it is, reproduces that arc clearly, and the much larger Wikidata layer agrees for the modern period.

The other story is about the record. The early innovations were made by peoples, and the pages that describe them hedge, list alternatives and give ranges; the modern entries are attributed to individuals with patents and birthplaces. When the record switches from archaeology to biography, the map switches with it, and the regions whose inventors were not written up in English, or whose innovations were collective, fade. A map of where invention was recorded in English is not the same as a map of where people invented.

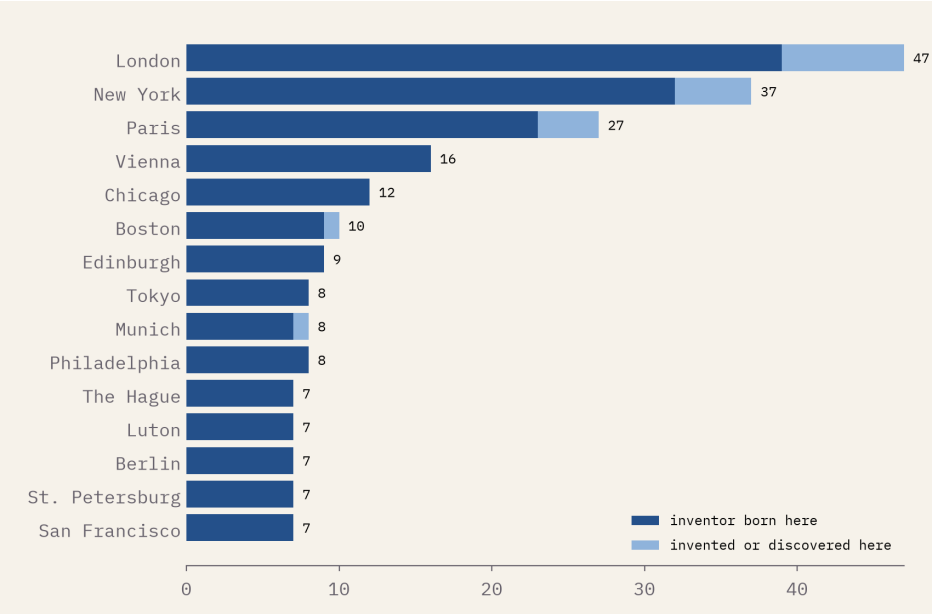


Figure 6: City areas with the most Wikidata inventions, split into inventors born there and inventions made or discovered there.

6 Limitations

The hand table is my reading of nine pages; other pages, and the specialist literature behind them, would add and move rows, and its coordinates are rough by design. The timeline is a curated list whose selection is itself a judgement, and about two in five of its modern entries were placed by hand at country level. The Wikidata filter is imperfect in both directions, especially among items with few sitelinks, and its dates are sparse. The inventor rule places many inventions at birthplaces, which can be far from where the work was done, and picks one inventor when there are several. Shares by period rest on small numbers before 1500. Population is today's and is only a rough yardstick for the past. Present-day borders are used throughout.

7 Conclusion

Put back where they came from, the world's inventions trace a centre that moves from East Africa to West Asia, China, Europe and North America, after a beginning in which farming, writing and zero were each invented more than once. The open record of that history is far more European and American than history itself, and far more attached to individual names. Both facts are worth seeing on one map.

Data and code availability

All data, code and figures are at https://github.com/Tristan-Hotham/inventions-map and archived at https://doi.org/10.5281/zenodo.23856736. The interactive atlas is at https://maps.tristanhotham.com/inventions-map/. The two tables built from Wikipedia text (early innovations and the timeline since 1500) are CC BY-SA 4.0; the other derived data, text and figures are CC BY 4.0, and the code is MIT.

AI statement

I wrote the text, with drafting and editing assistance from Claude (Anthropic), which also helped prepare the analysis code. I reviewed and checked all text and results.

Appendix: the early innovations

Sources: TH Timeline of historic inventions; TP Timeline of prehistory; NR Neolithic Revolution; DO Domestication; HW History of writing; WH Wheel; BA Bronze Age; IN List of Indian inventions and discoveries; IG Islamic Golden Age. The quoted passage for every row is in `data/early_innovations.csv`.

Table 5: Early innovations, places and dates as the pages give them.

Innovation	Place	Date	Kind	Page
Stone tools	Kenya	3.3 million years ago to 2.6 million years ago	finds	TH
Stone tools	Ethiopia (Oldowan tools)	3.3 million years ago to 2.6 million years ago	finds	TH
Hand axes (Acheulean stone tools)	Kenya	1.8 million years ago	single	TH
Control of fire	Africa (the page names Homo erectus but no place)	1.5 million years ago	single	TH
Cooking	Middle East	1,000,000 years ago to 778,000 years ago	single	TH
Hafting (fixing a stone point to a handle)	South Africa	498,000 years ago	single	TH
Wooden spears	Schöningen, Germany	335,000 years ago to 298,000 years ago	single	TH
Glue (birch tar)	Central Italy	198,000 years ago	single	TH
Clothing	Africa (continent only)	168,000 years ago to 88,000 years ago	single	TH
Harpoons	Democratic Republic of the Congo	88,000 years ago	single	TH
Bow and arrow	Sibudu Cave, South Africa	68,000 years ago to 58,000 years ago	single	TH
Sewing needle	Sibudu Cave, South Africa	68,000 years ago to 58,000 years ago	finds	TH
Sewing needle	Denisova Cave, Altai, Siberia	48,000 years ago	finds	TP
Ground stone axe	Australia	47,000 years ago to 43,000 years ago	single	TH
Mining (for red ochre)	Eswatini	45,000 years ago	single	TH
Tally sticks (Lebombo bone)	Eswatini	42,000 years ago to 40,000 years ago	single	TH
Flute	Germany	40,000 years ago	single	TH
Fired clay	Dolní Věstonice, Czech Republic (figurines)	26,000 years ago to 22,000 years ago	finds	TP
Fired clay	Xianren Cave, China (pottery vessels)	18050 BC to 17050 BC	finds	TP
Spear-thrower (atlatl)	France	15550 BC	single	TH
Domestication of the dog	Siberia	21,000 years ago	finds	TH
Domestication of the dog	Bonn-Oberkassel, Germany (oldest agreed remains)	12250 BC	finds	TP
Fish hook	Okinawa, Japan	20,000 years ago	single	TH
Bread	Jordan	12550 BC	single	TH
Spindle whorl	Nahal Ein Gev II, Israel	10050 BC	single	TH
Farming	Fertile Crescent (wheat, barley, lentils)	10000 BC to 8000 BC	independent	NR
Farming	Yangtze basin, China (rice)	11550 BC to 6250 BC	independent	NR
Farming	Yellow River basin, China (millet)	6050 BC to 5550 BC	independent	NR
Farming	Kuk Swamp, New Guinea Highlands (taro, bananas)	8000 BC to 7000 BC	independent	NR
Farming	Southern Mexico (maize, squash, beans)	7000 BC to 6000 BC	independent	NR
Farming	Southwestern Amazon basin (cassava)	8900 BC to 8300 BC	independent	TP

Table 5: Early innovations, places and dates as the pages give them.

Innovation	Place	Date	Kind	Page
Farming	Lake Titicaca, Andes (potatoes)	8000 BC to 5000 BC	independent	TH
Farming	Ethiopian highlands (coffee, teff, ensete)	3050 BC to 2050 BC	independent	NR
Farming	Sahel (sorghum, pearl millet)	3050 BC to 2050 BC	independent	NR
Farming	West Africa (African rice, yams, oil palm)	3050 BC to 2050 BC	independent	NR
Farming	Eastern North America (sunflower, sumpweed, goosefoot)	2500 BC to 1050 BC	independent	NR
Livestock (goats, sheep, pigs, cattle)	Fertile Crescent (goats, sheep, pigs, taurine cattle)	9050 BC to 8050 BC	independent	DO
Livestock (goats, sheep, pigs, cattle)	Baluchistan, Pakistan (humped zebu cattle)	7050 BC to 6050 BC	independent	DO
Livestock (goats, sheep, pigs, cattle)	East Asia (pigs)	6050 BC	independent	DO
Domestication of the cat	Fertile Crescent	8050 BC	single	DO
Alcohol fermentation (mead)	Southwest Asia	10000 BC to 9000 BC	single	TH
Mudbricks	Jericho	9000 BC	single	TH
Water well	Cyprus	8400 BC	single	TH
Boat (dugout canoe)	Pesse, Netherlands	8040 BC to 7510 BC	single	TH
Dog sled	Siberia	7000 BC	single	TH
Copper working	Great Lakes, North America (Old Copper Complex tools)	7500 BC to 6500 BC	finds	TH
Copper working	Pločnik and Belovode, Serbia (copper smelting)	5500 BC	finds	TP
Lead smelting	Çatalhöyük, Turkey	6500 BC	single	TH
Kiln	Mesopotamia	6000 BC	single	TH
Irrigation	Khuzistan, Iran	6000 BC to 5001 BC	single	TH
Cheese	Poland	5500 BC to 5200 BC	finds	TH
Cheese	Dalmatian coast, Croatia	5500 BC to 5200 BC	finds	TH
Sailing	Mesopotamia	5500 BC	single	TH
Cotton thread	Mehrgarh, Indus Valley	5000 BC	single	TH
Lost-wax casting	Palestine	4500 BC to 3500 BC	candidates	TH
Lost-wax casting	Indus Valley	4500 BC to 3500 BC	candidates	TH
Paved roads	Ur, Mesopotamia	4000 BC	single	TH
Plumbing (clay pipes)	Nippur, Mesopotamia	4000 BC	single	TH
Lock	Nineveh, Assyria	4000 BC	single	TH
Wheel and wagon	North of the Black Sea (miniature clay wheels)	4100 BC to 4000 BC	finds	WH
Wheel and wagon	Uruk, Mesopotamia (wagon pictographs)	3500 BC to 3350 BC	finds	WH
Wheel and wagon	Bronocice, Poland (wagon on a pot)	3631 BC to 3380 BC	finds	WH
Wheel and wagon	Northern Germany (wheel ruts)	3400 BC	finds	WH
Wheel and wagon	Ljubljana Marshes, Slovenia (oldest wooden wheel)	3130 BC	finds	WH
Potter's wheel	Ur, Mesopotamia	4000 BC to 3100 BC	single	WH
Spoked wheel and chariot	Sintashta culture, southern Urals	2000 BC	single	WH
Domestication of the horse	Eurasian steppes	3500 BC	single	TH
Silk	China	3630 BC	single	TH
Plough	Bubeneč, Czech Republic	3500 BC	finds	TH
Plough	Kalibangan, Indus Valley	2800 BC	finds	TH
Bronze	Pločnik, Serbia (tin bronze foil, dating disputed)	4650 BC	independent	BA
Bronze	Maykop culture, North Caucasus (arsenical bronze)	3600 BC to 3400 BC	independent	BA

Table 5: Early innovations, places and dates as the pages give them.

Innovation	Place	Date	Kind	Page
Bronze	Sumer, Mesopotamia	3600 BC to 3400 BC	independent	BA
Writing	Uruk, Mesopotamia (cuneiform)	3400 BC to 3100 BC	independent	HW
Writing	Abydos, Egypt (hieroglyphs)	3400 BC to 3100 BC	independent	HW
Writing	Late Shang China (oracle bone script)	1250 BC to 1050 BC	independent	HW
Writing	Maya region, Mesoamerica (Maya script)	300 BC to 200 BC	independent	HW
Sword	Arslantepe, Turkey	3300 BC	single	TH
Papyrus	Egypt	3000 BC to 2560 BC	single	TH
Dice	Burnt City, Iran	2800 BC to 2500 BC	single	TH
Prosthesis	Ancient Egypt	3000 BC to 2800 BC	finds	TH
Prosthesis	Iran (artificial eye)	3000 BC to 2800 BC	finds	TH
Rhinoplasty	Egypt	3000 BC to 2500 BC	single	TH
Ruler (measuring rod)	Nippur, Mesopotamia	2650 BC	single	TH
Public sewers and sanitation	Mohenjo-daro, Indus Valley	2600 BC	single	TH
Balance weights and scales	Egypt (Fourth Dynasty)	2600 BC	single	TH
Fork	Qijia culture, China	2400 BC	single	TH
Iron smelting	Kaman-Kalehöyük, Turkey	2200 BC to 2000 BC	single	TH
Water clock	Babylonia	2000 BC to 1600 BC	candidates	TH
Water clock	Mohenjo-daro, Indus Valley	2000 BC	candidates	TH
Scissors	Mesopotamia	2000 BC	single	TH
Noodles	Lajia, China	2000 BC	single	TH
Alphabet (Proto-Sinaitic script)	Sinai, Egypt	1850 BC to 1800 BC	single	HW
Phoenician alphabet	Phoenicia (Lebanon)	1050 BC	single	HW
Greek alphabet, the first with vowels	Greece	800 BC	single	HW
Sundial	Ancient Egypt	1500 BC	candidates	TH
Sundial	Babylonia	1500 BC	candidates	TH
Glass	Mesopotamia	1500 BC	candidates	TH
Glass	Ancient Egypt	1500 BC	candidates	TH
Seed drill	Babylonia	1500 BC	single	TH
Rubber	Mesoamerica (the Mesoamerican ballgame)	1400 BC	single	TH
Concrete	Tiryns, Greece	1400 BC to 1200 BC	single	TH
Distillation	Mesopotamia (Akkadian tablets)	1200 BC	single	TH
Saddle	Assyria	700 BC	single	TH
Crossbow	China	650 BC	single	TH
Coins	Phoenicia (Lebanon)	600 BC	candidates	TH
Coins	Lydia (Sardis, Turkey)	600 BC	candidates	TH
Wagonway (the Diolkos)	Isthmus of Corinth, Greece	600 BC	single	TH
Crank (rotary quern)	Carthage	550 BC to 500 BC	candidates	TH
Crank (rotary quern)	Celtiberian Spain	500 BC to 400 BC	candidates	TH
High-carbon (Wootz) steel	South India	600 BC to 300 BC	single	TH
Systematic surgery (cataract surgery, plastic surgery)	Northern India (Sushruta Samhita)	600 BC to 200 BC	single	TH
Cast iron	Luhe County, Jiangsu, China	500 BC to 401 BC	single	TH
Catapult	Magadha, India	485 BC	single	TH
Camera obscura	China (described by Mozi)	400 BC	single	TH
Zero	Babylon	400 BC to 300 BC	independent	TH
Zero	Maya region, Mesoamerica	AD 1 to AD 300	independent	TH
Zero	India (Brahmagupta)	AD 628	independent	IN
Decimal place-value numerals	Indian subcontinent	AD 1 to AD 600	single	IN

Table 5: Early innovations, places and dates as the pages give them.

Innovation	Place	Date	Kind	Page
Traction trebuchet	China	400 BC to 301 BC	single	TH
Gears	China	400 BC to 301 BC	single	TH
Nailed horseshoe	Etruria, Italy	400 BC to 301 BC	single	TH
Analogue computer (Antikythera mechanism)	Rhodes, Greece (possibly)	300 BC to 201 BC	single	TH
Archimedes' screw	Nile, Egypt	300 BC to 201 BC	single	TH
Canal lock	Canal of the Pharaohs, Egypt	283 BC to 246 BC	single	TH
Water wheel	India (Pali texts)	400 BC to 301 BC	candidates	TH
Water wheel	Persia	350 BC	candidates	TH
Water wheel	Hellenistic world (Philo of Byzantium)	280 BC to 220 BC	candidates	TH
Blast furnace	China	300 BC to 101 BC	single	TH
Paper	Han dynasty China	200 BC to AD 200	single	TH+HW
Compass	Han dynasty China	206 BC	single	TH
Astrolabe	Perga (Apollonius of Perga)	200 BC to 150 BC	single	TH
Glass blowing	Jerusalem	37 BC to 14 BC	single	TH
Aeolipile (simple steam turbine)	Alexandria (Hero of Alexandria)	AD 1 to AD 100	single	TH
Seismometer	Han dynasty China (Zhang Heng)	AD 132	single	TH
Woodblock printing	Han dynasty China	AD 100 to AD 220	single	TH
Crystallised sugar	India	AD 1 to AD 300	single	IN
Suspension bridge	Pre-Columbian South America	AD 301 to AD 400	independent	TH
Suspension bridge	Hindu Kush (Afghanistan and Pakistan)	AD 301 to AD 400	independent	TH
Horse collar	Northern and Southern dynasties China	AD 401 to AD 500	single	TH
Spinning wheel (charkha)	Maharashtra, India	AD 500 to 1000	single	TH
Toilet paper	Sui dynasty China	AD 589	single	TH
Toothbrush	Tang dynasty China	AD 619	single	TH
Greek fire	Constantinople	AD 672	single	TH
Banknote	Tang dynasty China	AD 618 to AD 700	single	TH
Porcelain	Northern China	AD 618 to AD 700	single	TH
Gunpowder	Tang dynasty China	AD 801 to AD 900	single	TH
Playing cards	Tang dynasty China	AD 801 to AD 900	single	TH
Algebra (al-Khwarizmi's text) and Indian arithmetic in the Islamic world	Baghdad	AD 825 to AD 830	single	IG+IN
Fire lance (the first firearm)	Song dynasty China	AD 901 to 1000	single	TH
Fountain pen	Fatimid Egypt	AD 974	single	TH
Movable type	Song dynasty China (Bi Sheng)	1040 to 1088	single	TH
Hand cannon	Heilongjiang, China	1201 to 1300	single	TH
Eyeglasses	Italy	1286	single	TH
Printing press with movable type	Mainz, Germany (Johannes Gutenberg)	1439	single	TH

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